

ARP Spoofing Attack using Packet Tracer

Background

Design a Packet Tracer simulation to demonstrate an ARP spoofing attack observing how a malicious device can redirect network traffic intended for one device to another. Analyze detection methods and mitigation strategies for ARP spoofing. Share screenshots as evidence of your work.

Methodology

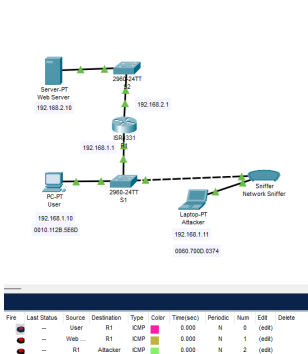
Topology and Devices Configuration

Device	IP Address	Mac Address
PC-PT (User)	192.168.1.10	0010.112B.5E6D
Laptop-PT (Attacker)	192.168.1.11	0060.700D.0373
Server-PT (Web Server)	192.168.2.10	0004.9A90.D7
R1 (S1 Default Gateway)	192.168.1.1	002.1638.A501
R1 (S2 Default Gateway)	192.168.2.1	002.1638.A501

Steps ([ARP Spoofing Packet Tracer File](#))

1. Set up the network with PC-PT (User), 2 Switches, a router (R1), and a web server, each with their respective IP and MAC addresses.
2. Check the ARP cache on PC1 using the "arp -a" command, to check/verify if the MAC address of the default gateway (R1) is known.
3. Test the connection to the remote web server from PC1 using ping 192.168.2.10
4. Verify that the MAC address of the default gateway (R1) is known on PC1's ARP cache using the "arp -a" command.
5. Add, configure and record the Attacker's Laptop.
6. Change the threat actor's MAC address to the default gateway's (R1) MAC address to intercept communications.
7. Add/Connect a Network Sniffer in between S1 and the Attacker's Laptop.

- ## Results

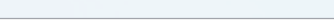
[illegible]

Step 8

Step 1-4

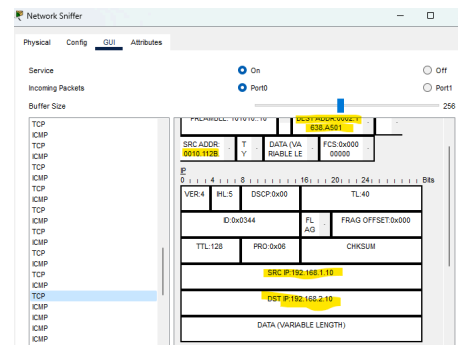
Step 5 and 7

```
Switch>en
Switch#show mac add
          Mac Address Table
-----
Vlan      Mac Address      Type      Ports
-----
1         0002.1638.a501   DYNAMIC   Fa0/2
1         0000.112b.5e6d   DYNAMIC   Fa0/1
Switch#
```



Step 11

Step 10



Step 12

This successful man-in-the-middle attack highlights the potential risks and unauthorized data access.

Conclusion

Detection Methods

1. **MAC Address and ARP Inspection:** Use ARP inspection on the network switch to validate ARP packets and drop suspicious ones.
2. **ARP Spoofing Detection Software:** Use tools like Wireshark or ARPwatch to detect and alert on ARP spoofing attacks.

Mitigation Strategies:

1. **Static ARP Entries:** Configure static ARP entries on critical devices to prevent ARP poisoning.
2. **Network Segmentation:** Segment the network into smaller subnets to limit the scope of ARP spoofing attacks.
3. **Encryption and Authentication:** Use encryption and strong authentication mechanisms to protect network traffic from being intercepted and manipulated.
4. **Periodic ARP Cache Clearing:** Periodically clear ARP caches on devices to remove potential spoofed entries.

References

CrowdStrike. (2024, January 22). *What is ARP Spoofing? How to Prevent & Protect* –

CrowdStrike. crowdstrike.com

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https://www.youtube.com/watch?v=R_kC-jCEzho

Psybernautic. (n.d.). GitHub - Psybernautic/ArpSpoofing: ARP Spoofing Detection and

Simulation with Python: Learn how to detect and defend against one of the most common attacks on local networks. GitHub. <https://github.com/Psybernautic/ArpSpoofing>